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1.Introduction

The **Smart Library Request Workflow** is a digital solution developed using ServiceNow to automate and streamline library services. It replaces traditional manual processes with an efficient, user-friendly system for requesting, tracking, and managing library resources.

This system enhances accessibility, reduces delays, and improves overall user experience for students, faculty, and library staff.

1.1 Purpose of the Project

The main purpose of this project is to:

- Simplify the book request process
- Automate approval and issue workflows
- Provide real-time tracking of requests
- Improve communication through notifications
- Reduce manual workload for librarians

Key Features

User Features

- Submit book requests (Issue / Reserve / Purchase)
- Track request status in real-time
- Receive notifications for updates

Librarian Features

- Manage and process requests
- Approve or reject new book requests
- Issue and track books

System Features

- Automated workflows using Flow Designer
- Role-based access control
- Dashboard and reporting

1.2 Objectives

- Automate library request handling
- Minimize processing time
- Ensure transparency in request tracking
- Enhance user satisfaction
- Improve operational efficiency

2. IDEATION PHASE

2.1 Problem Definition

Start by identifying what's broken or inefficient:

- Manual book requests (paper/email chaos)
- No real-time tracking of requests
- Delay in approvals (librarian/admin)
- No visibility of book availability
- Poor user experience for students

2.2 Stakeholders

- Students / Faculty (Requesters)
- Librarian (Approver & Manager)
- Admin (Policy control)
- IT Team (ServiceNow developers)

2.3 Brainstorming

Request Features

- Request book (physical / digital)
- Request new book purchase
- Reserve currently unavailable book
- Bulk requests for faculty

Smart Enhancements

- Auto-suggest books while typing
- Availability check (real-time)
- AI-based recommendation (based on history)
- Estimated wait time prediction

Workflow Automation

- Auto-routing to librarian
- Approval flow (if needed for purchase)
- Notifications (email / mobile)

3. REQUIREMENT ANALYSIS

3.1 Stakeholders & Their Needs

Students / Faculty

- Easy way to request books
- Check availability instantly
- Track request status
- Get notifications

Librarian

- View and manage all requests
- Approve/reject purchase requests
- Issue/assign books
- Monitor demand

Admin

- Define policies (limits, approvals)
- View reports and analytics

IT Team

- Maintain system
- Ensure workflow automation

3.2 Functional Requirements

User Requirements

- User can submit a book request via Service Catalog
- User can select request type:
    - Issue Book
    - Reserve Book
    - Request New Book
- User can view request status (Pending / Approved / Issued / Rejected)
- User receives notifications (email/app)

System Requirements

- System validates book availability
- System prevents duplicate requests
- System auto-assigns request to librarian
- System maintains request history
- System supports waitlist for unavailable books

Librarian Requirements

- Librarian can approve/reject requests
- Librarian can update request status
- Librarian can issue/return books
- Librarian can view dashboard (pending, completed requests**)**

Admin Requirements

- Admin can configure rules (max books per user)
- Admin can generate reports
- Admin can manage book inventory (optional integration)

3.3 Non-Functional Requirements

Security

- Role-based access (User / Librarian / Admin)
- Data privacy protection
- **Performance**
- Request processing within seconds
- Support multiple users simultaneously

Usability

- Simple UI (mobile-friendly)
- Minimal steps to submit request

Scalability

- Should handle increasing number of users and requests

3.4 Workflow Requirements

Standard Flow

1. User submits request
2. System checks availability
3. Route to librarian
4. Librarian approves/rejects
5. Book issued / waitlisted
6. User notified
7. Request closed

Alternate Flows

- Book not available → Add to waitlist
- New book request → Approval required
- Duplicate request → System blocks or warns

4. PROJECT DESIGN

4.1 ServiceNow Implementation Design

Service Catalog

- Create Catalog Item: “Library Request”
- Variables:
    - Book Name
    - Request Type
    - Priority
    - **Flow Designer**

Flow Steps:

1. Trigger → Catalog Item Request
2. Check Condition (Availability)
3. Decision:
    - Available → Update status = Issued
    - Not Available → Waitlist
    - New Book → Approval Action
4. Notify User
5. Close Request

Business Rules

- Prevent duplicate requests
- Auto-fill user details
- Validate inputs

Notifications

- **Email templates:**
    - Request Confirmation
    - Approval/Rejection
    - Issue Notification

4.2 Security Design

Role-based access:

- User → Submit/view requests
- Librarian → Manage requests
- Admin → Full control

Access Control Rules (ACL):

- Restrict table access
- Protect sensitive data

4.3 UI Design

Service Portal Pages

Request Form

- Book Name (search field)
- Request Type (dropdown)
- Description (optional)

My Requests Page

- View status
- Track progress

Librarian Dashboard

- Pending approvals
- Request list
- Filters (status/type)

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Project Phases

Phase 1: Requirement Analysis

- Gather user needs
- Define functional requirements

Phase 2: Design

- System architecture
- Database & workflow design

Phase 3: Development

- Create Service Catalog
- Build Flow Designer workflows
- Configure tables

Phase 4: Testing

- Unit Testing
- Integration Testing
- User Acceptance Testing (UAT)

Phase 5: Deployment

- Move to production
- User training

Phase 6: Maintenance

- Bug fixes
- Feature updates

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

1. Response Time

- Time taken to submit request
- Time to update status

🔹 **2. Load Testing**

- Multiple users submitting requests simultaneously
- System handles concurrent request

🔹 **3. Workflow Execution Time**

- Time taken for Flow Designer to process request

🔹 **4. Scalability**

- Increasing number of requests over time
- System performance remains stable

8. CONCLUSION

The Library Management Project successfully optimized the process of borrowing and managing books within ServiceNow by implementing structured workflows, role-based access control, and automated approvals. Students can submit borrow requests seamlessly, while librarians can review, validate, and process requests efficiently through controlled access and defined responsibilities.

The system enhances transparency by incorporating request status tracking, automated notifications, and reporting capabilities, ensuring that all stakeholders remain informed throughout the process. Proper validations reduce errors, prevent unauthorized actions, and maintain accurate record management.

Overall, the solution improves operational efficiency, strengthens accountability, and provides a scalable framework that can be extended to support additional academic services such as digital resources, reservation systems, or asset tracking in the future.

9. APPENDIX

GitHub Link :

https://github.com/jonesruok18-cyber/Smart-Library-Request-Workfllow-in-ServiceNow